Deep Learning 1

Git, GitHub & Classroom 50

How you get, submit and get feedback on your assignments

Andrei Vlad Nicula

University of Amsterdam
Informatics Institute

Contents

Git 3
GitHub 12
Classroom 50 18

Git

Sound familiar?



`assignment1/`



`assignment1_copy/`



`assignment1_final/`



`assignment1_final_final/`

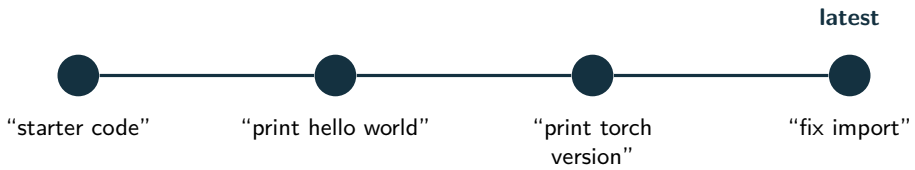


`assignment1_final_REAL/`

- Which one did you hand in?
- What changed between `final` and `final_final`?

What is Git?

Git is a version control system: it keeps the full history of a folder, so you need only *one* folder.



- Every dot is a **commit**: a snapshot of all your files plus a short message.
- You can look at, compare or go back to any commit.
- A folder tracked by Git is called a **repository** (repo).

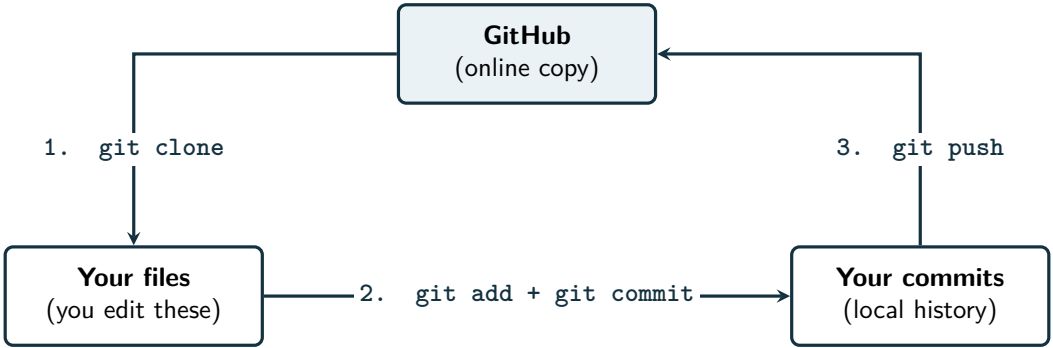
Set up Git (once)

Install Git from https://git-scm.com/downloads, then configure your name and email:

```
git --version          # check that it is installed
git config --global user.name  "Your Name"
git config --global user.email "you@example.com"
```

Use the same email as your GitHub account, so your commits are linked to you.

The only workflow you need



Clone once. Then repeat: edit, add, commit, push.

1. `git clone`: download your repository

Makes a local copy of a repository from GitHub, with its full history. For example:

```
git clone https://github.com/UvA-MSc-AI/deep-learning-1-assignment-<N>-<username>.git
cd deep-learning-1-assignment-<N>-<username>
```

- Replace <N> with the assignment number and <username> with your GitHub username.
- You only do this once per assignment.
- The next section shows where to copy this URL from.

2. `git add` + `git commit`: save a snapshot

Edit `main.py`, then:

```
git status                    # what changed?
git add main.py               # choose what goes into the snapshot
git commit -m "Print hello world and the torch version"
```

- `git add .` adds every changed file.
- The message says *what* you changed. Commit small and often.
- A commit only exists on your computer until you push it.

3. `git push`: upload your commits

```
git push
```

- Sends your new commits to GitHub.
- **Every push is a submission**: it starts the autograder.
- Working on two machines (laptop and Snellius)? Run `git pull` first to fetch the commits you pushed from the other one.

Git: documentation

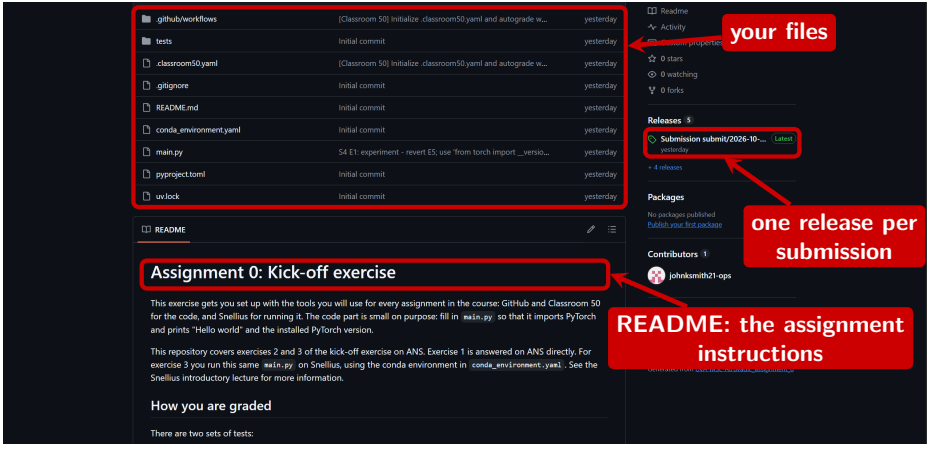
- Download Git: https://git-scm.com/downloads
- Pro Git, the free official book: https://git-scm.com/book/en/v2
- Reference for every command: https://git-scm.com/docs
- One-page cheat sheet:
  https://education.github.com/git-cheat-sheet-education.pdf

GitHub

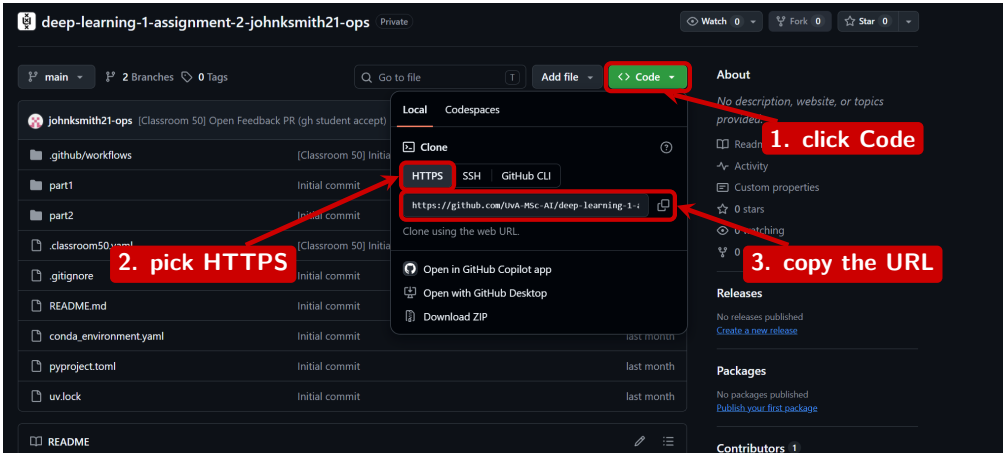
What is GitHub?

- A website that stores Git repositories online.
- **Git** is the tool on your computer. **GitHub** is the place you push to.
- Your work is backed up, and we can see and grade it.
- You need a (free) GitHub account: `https://github.com/signup`

A repository on GitHub



Copy the clone URL



Then: `git clone <paste the URL here>`

Logging in from the terminal

The first clone or push asks you to log in to GitHub. The simplest way, on every operating system, is the GitHub CLI (https://cli.github.com):

```
gh auth login      # GitHub.com -> HTTPS -> yes -> login with a web browser
```

- Git then uses this login for every HTTPS clone and push.
- Your GitHub *password* does not work in the terminal.
- Prefer SSH keys? That works too: pick SSH in the Code menu instead.

GitHub: documentation

- Create an account: https://docs.github.com/en/get-started/start-your-journey/creating-an-account-on-github
- Hello World tutorial: https://docs.github.com/en/get-started/start-your-journey/hello-world
- Cloning a repository: https://docs.github.com/en/repositories/creating-and-managing-repositories/cloning-a-repository
- Logging in with the GitHub CLI: https://cli.github.com/manual/gh_auth_login
- Using SSH keys: https://docs.github.com/en/authentication/connecting-to-github-with-ssh

Classroom 50

What is Classroom 50?

- The tool we use to hand out and grade the coding assignments.
- For each assignment you get your **own private repository**, with the starter code already in it:
  `deep-learning-1-assignment-<N>-<username>`
- Only you and the course staff can see it.
- Every push is autograded, and the feedback appears on GitHub.

Step 1: join the course organisation

- Today you will get an **email from GitHub** on your **UvA email** address, inviting you to the `UvA-MSc-AI` organisation.
- Open it and accept, while logged in to your GitHub account. Any GitHub account works; it does not need to use your UvA email.
- The invitation **expires after 7 days**.
- No email? Check your spam folder, then contact us ED Discussion.

Step 2: accept the assignment

- Each assignment has its own link, posted in a **Canvas announcement**.
- Open it, **Sign in with GitHub**, click **Accept assignment**.
- Your repository is created; then `git clone` it.

Out now: assignments 0 and 1

Assignment 0: Kick-off

Deadline: **Thursday 29 October, 23:59**



https://classroom50.org/UvA-MSc-AI/deep-learning-1/assignments/assignment-0/accept?k=3tx61ev0

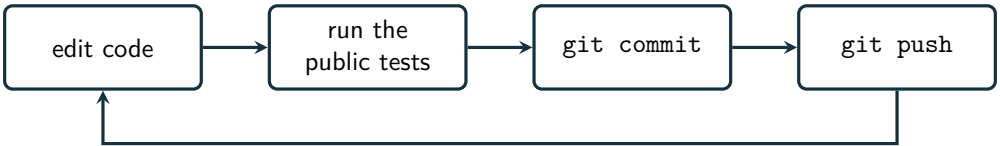
Assignment 1: MLPs & Backprop

Deadline: **Monday 9 November, 23:59**



https://classroom50.org/UvA-MSc-AI/deep-learning-1/assignments/assignment-1/accept?k=3tx61ev0

Step 3: work, commit, push



- The README of each assignment says how to run the public tests locally.
- It is *highly recommended* to implement your own tests, so you can check your code before pushing.
- Push as often as you like: each push is graded again.

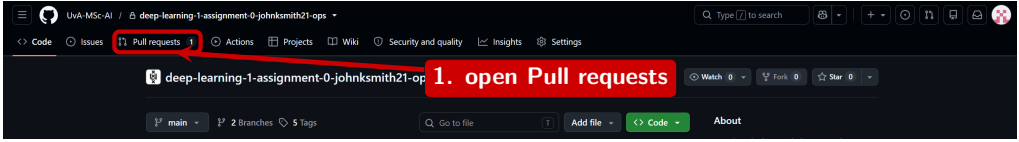
What happens when you push

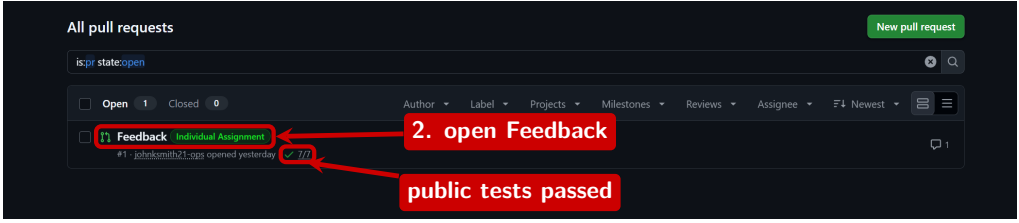
1. **Public tests** and the **linter** (code style) run first. You have these tests too, so nothing here is a surprise.
2. Then the **graded tests** run. These are hidden, and **they decide your grade**.
3. The result is posted as a comment in your **Feedback** pull request.

When is the comment updated?

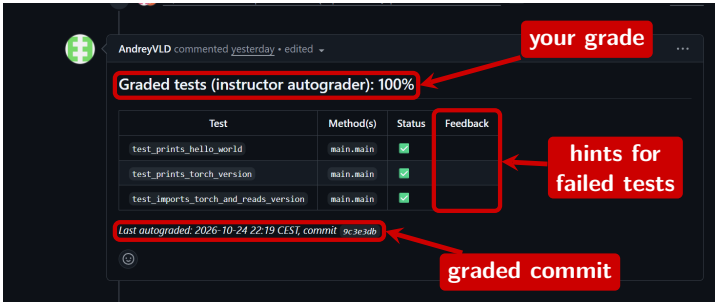
Every 15 minutes during the day and every hour at night, the comment is updated with the feedback for your latest push.

Step 4: read your feedback





Step 4: read your feedback



- **3. The comment**: score, one row per graded test.
- A failed test shows a hint in the Feedback column.
- Check the commit: it should be your latest push.

Classroom 50: documentation

- Student guide: https://github.com/foundation50/classroom50/wiki/Web-Student-Guide
- About pull requests: https://docs.github.com/en/pull-requests/reference/pull-requests
- Assignment links: Canvas announcements

Today

1. Create a GitHub account.
2. Install Git and run `gh auth login`.
3. Accept the GitHub invitation in your UvA email.
4. Accept the assignments, then clone them.
5. Solve, commit, push.
6. Check your Feedback pull request.

Links

- **Git:** `https://git-scm.com/book/en/v2`
- **GitHub:** `https://docs.github.com/en/get-started`
- **Classroom 50:** `https://github.com/foundation50/classroom50/wiki/Web-Student-Guide`
- **Assignment links: Canvas announcements**